Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Efficient and Scalable Production of EUV Photoresist Precursors by AI-Driven Flow Chemistry**			
Korea's 12 National Strategic Technology Fields	**Semiconductor/Display (√)**, Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), **AI (√)**, Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical & Electronics		***Chemicals***	
	Medium Category	Small Category	Medium Category	Small Category
	Semiconductor Devices and Systems	***Semiconductor Materials***	***Fine Chemicals***	***Photosensitive Materials***
Concept of Proposed Technology	Extreme Ultraviolet (EUV) lithography is a critical step in advanced chip manufacturing, with increasing demands for higher resolution driven by the rapid expansion of AI computing. To address these requirements, new resist chemistries are being developed (1), creating a need for resource-efficient, cost-effective, and safe production of novel EUV photoresist precursor materials using innovative process technologies. Furthermore, production technologies must enable rapid and flexible scaling to meet dynamic market demands. A production technology for EUV photoresist precursor materials using AI-supported Flow Chemistry is proposed. Flow Chemistry enables the transition from batch to continuous processing, providing improved safety, efficiency, and simplified scale-up in industrial chemical production. The integration of AI methods and tools across chemical process development and production stages can significantly improve process efficiency while lowering production costs. This proposal presents			

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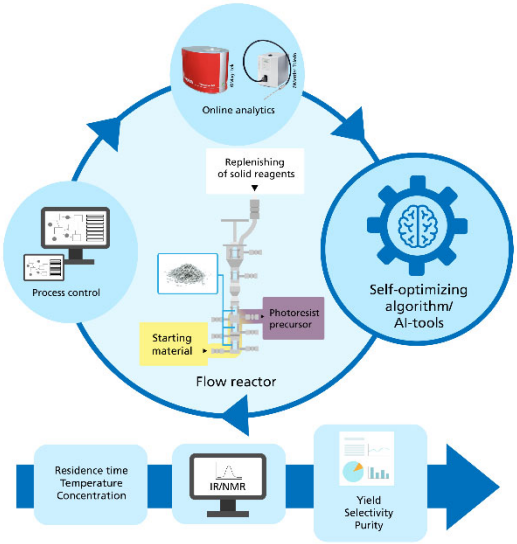
	the first-time application of Flow Chemistry tailored to the synthesis of novel EUV photoresist precursors, alongside the concurrent use of AI to enhance production efficiency.  *Illustration of project idea "Efficient and Scalable Production of EUV Photoresist Precursors by AI-Driven Flow Chemistry"*
Rationale for Joint R&D Project	The synthesis of novel EUV photoresist precursors, particularly metal-containing precursors, presents significant challenges in terms of process safety, efficiency (cost, yield, productivity), and scalability. Addressing these challenges requires the development of continuous synthesis processes, supported by AI-based methods for automated optimization and accelerated process development. Fraunhofer IMM brings in its expertise in Flow Chemistry and its experience in implementing AI methods and tools across chemical process development and production. The Korean partner contributes complementary chemical and application know-how in the field of novel precursor materials for EUV photoresists. The collaboration is essential to adapt Flow Chemistry technology and AI methods to the specific requirements of EUV photoresist precursors. By combining the partners' expertise, the project aims to reduce development time and associated costs while ensuring a reliable and efficient process. A medium-scale demonstration within the project will validate the approach and demonstrate its feasibility for industrial implementation.
Key Words	1. EUV photoresist precursors 2. AI-driven Flow Chemistry 3. Flow Chemistry for process safety, efficiency, and scalability

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 4	End of the Project	Level 7

R&D Objectives

The objective of this project is to develop an efficient manufacturing process based on AI-driven Flow Chemistry for dry-lithography-compatible high-NA EUV photoresist precursors that enable next-generation semiconductor patterning. The target EUV photoresist is designed to achieve a film thickness below 20 nm, a resolution better than 8 nm, line-edge roughness (LER) below 1.2 nm, photosensitivity under 60 mJ/cm², and etch resistance greater than 10, thereby meeting the stringent requirements for advanced chip manufacturing for artificial intelligence (AI), data centers, and high-performance computing.

To achieve this objective, the project will focus on the following R&D goals:

1. **Development of ultra-high pure EUV photoresist precursors synthesis using green chemistry principles:** A continuous-flow (Flow Chemistry) synthesis platform will be implemented to minimize human toxicity and environmental impact while producing high-quality precursors.
2. **Establishment of an AI-driven synthesis process optimization system:** This system will accelerate the development of ultra-high pure EUV photoresist precursors by enabling data-driven control of reaction conditions and material properties, ensuring consistent quality and process efficiency.

R&D Target Specifications

This project aims to develop EUV photoresist precursors for EUV lithography targeting for a resolution below 8 nm, line-edge roughness (LER) under 1.2 nm, and sensitivity below 60 mJ/cm², leveraging AI-driven Flow Chemistry. The following material quality specifications for the photoresist precursors must be met:

Items	Target Spec.	Analytical method
Appearance	Colorless transparent liquid	-
Purity	>99%	GC
Metal impurities	<100ppb	ICP-MS

Aside these material specifications to be met, the following R&D targets will be followed:

- Successful transfer of the current batch procedure of precursor material synthesis with its limitations to an efficient continuous procedure exploiting the process benefits of Flow Chemistry.
- Implementation of AI tools in the corresponding process development phase in the lab scale to accelerate process optimization.
- Adjusted Flow Chemistry reactor technology to address the specific challenges of precursor synthesis in larger scale.
- Plant concept validated by a demonstration plant at relevant scale equipped with AI tools to assure process efficiency. Validation criteria:
  - Quality of produced precursors in spec for use in industrial photoresist production (see

	table above) ○ Throughput achieved reaches relevant demonstrator scale for industry ○ Throughput variability in the range of a factor of ten (to address changing / growing market needs) ○ Existing concept for a further increase of throughput by a further factor of ten ○ AI tool implementation to secure process efficiency
R&D Plan	The following project plan assumes an overall project duration of 36 months as stated in the call for proposals. Nevertheless, the ambition would be - if in agreement with the call requirements - to reduce project duration to 30 months to accelerate the implementation of the technology in industry. Work package (WP) 1 (Project month (M) 1 to 6) represents Phase I (Pre-study) from the call for proposals. The following work packages then are part of Phase II (Main R&D). **WP1 Development of basic continuous process for EUV photoresist precursor synthesis and corresponding process scheme (M1-M6) – Fraunhofer IMM, Korean SME** The synthesis of novel EUV photoresist precursors involve the use of metal containing reagents and metal organic reagents and include the need for solids handling in the chemicals process. This makes batch processes in this case extremely challenging due to hazards such as explosiveness and flammability. Furthermore, the precursors for photoresists require exceptionally high quality, particularly in terms of purity and metal ion content. These two aspects, make the advantages of Flow Chemistry particularly compelling. WP1 is dedicated to transfer the current batch procedure recipe into a continuous procedure. A corresponding lab set-up will be established including flow reactor technology from Fraunhofer IMM dedicated to solids handling and metal reagent synthesis. Lab experiments will be performed to show the general feasibility of running the processes in continuous manner / Flow Chemistry approach. Selected samples from the lab experiments will be inspected with high quality analysis by the Korean SME to monitor basic alignment with the quality requirements of the precursor material. Based on the results, a general process scheme is elaborated as basis for further optimization studies (WP2) but also for preparing the design of a demonstration plant (WP4). ***Milestone 1:*** Proof of concept for Flow Chemistry process for EUV photoresist precursors. (M6) **WP2 Implementation of AI tools for process optimization at lab scale & process optimization (M7-18) – Fraunhofer IMM, Korean SME** The lab set-up of WP1 will be further developed to allow for an automated operation. Fraunhofer IMM will link AI tools to the automated lab plant to exploit self-optimization in the following lab experiments for further process improvement. Fraunhofer IMM will use AI tools from its portfolio including commercially available solutions but also own algorithms and will further develop them to adjust them to the specific process. ***Milestone 2:*** Via AI-tools optimized lab process for the EUV photoresist precursor validated (M15).

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	WP3 Adjustment of core flow reactor technology to specific process needs (M10-21) - Fraunhofer IMM The handling of solids in the chemical process and the synthesis of metal organic reagents are essential in the considered synthesis of the EUV photoresist precursors. Fraunhofer IMM will bring its flow reactor technology. Based on the learnings in WP1 and WP2, the flow reactor technology will be further developed to match best process needs. A further developed lab reactor will be realized for implementation in the lab experiments of WP2. Furthermore, the reactor will be further developed for the demonstration scale and realized for the implementation in the demonstrator plant of WP4. ***Milestone 3:*** Further developed reactor technology realized and ready for testing upfront implementation in the demonstrator plant (M21). **WP4 Demo plant with AI implementation – realization, testing and validation (M18-36) – Korean SME, Fraunhofer IMM,** The general process scheme of WP1 will be further developed as basis for the realization of a demonstrator plant at industrial relevant throughput range. The core Flow Chemistry parts of the demonstrator plant will be developed, assembled and pre-tested at Fraunhofer IMM. Process automatization will be developed. Based on the experience with the AI tools during lab experiments, these will be further developed for implementation in the demonstrator plant. Aside optimization, these AI tools target at a controlled and efficient operation of the plant even under process fluctuations. The Korean SME will prepare on site, the erection of the demonstrator plant. After pretesting of central elements at Fraunhofer IMM, these parts will be transferred to the Korean SME. The complete assembly will then be operated by the Korean SME to validate core aspects of the development (like product quality, throughput range, AI tool efficiency, etc.). The activities within WP4 will also include considerations on further scale-up of the demonstrator plant capacity. ***Milestone 4:*** Piping and instrumentation document for demonstrator plant completed (M21). ***Milestone 5:*** Delivery of Flow Chemistry elements for the demonstrator plant to the SME (M27) ***Milestone 6:*** AI tools connected to process automation system of the demonstrator plant (M27) ***Milestone 7:*** Demonstrator plant ready for validation runs (M30)

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Fraunhofer IMM has a more than 20-years long track of experience in developing, realizing and applying microstructured reactors and flow reactors (Flow Chemistry) to improve industrial chemical production processes. In context of the suggested project it is of special importance that Fraunhofer IMM set a focus on developing flow reactor technology for the synthesis of metal organic reagents (like Grignard reagents and Zn organic compounds) which involves solids handling in flow and replenishing of used solids in course of the process to keep it continuous. Furthermore, Fraunhofer IMM has built up experience in the field of industrial process automatization, the development of self-optimizing lab plants, and in this context developing and using corresponding algorithms and AI tools. In the project Fraunhofer IMM will bring in this technology basis and further develop it adjusted to the specific task. To our knowledge it is the first time application of Flow Chemistry combined with AI tools in the context of EUV photoresist precursor synthesis. Fraunhofer IMM will work jointly with the Korean SME to transfer the given batch procedure in a continuous one exploiting the benefits of Flow Chemistry in terms of product quality, safety, efficiency, and scalability. Fraunhofer IMM will establish an automated lab set-up, link it with AI tools and perform optimization experiments, and will further develop its reactor technology to demonstrator scale. Jointly with the Korean SME, Fraunhofer IMM will develop a demonstration plant equipped with an AI assisted process automatization for operation at the site of the Korean SME. Flow chemistry has gained increasing importance, particularly in the last 10-15 years. It is now widely used in both academic research and industrial applications, mainly within the chemical-pharmaceutical industry. The advantages of this technology, especially for exothermic and high-risk syntheses – as in the synthesis and use of metal inorganic and organic compounds, are undisputed and are based on improved temperature management and an inherent increase in safety due to relatively small reactor volumes, even in pilot-scale plants. Conventional batch synthesis linked to metal inorganic and organic compounds has several disadvantages, such as e.g. variable incubation times until the reaction starts, a generally high exothermicity requiring efficient temperature management, and significant sensitivity of the reagents to water and oxygen. The technology developed by Fraunhofer IMM addresses these challenges, overcoming them in handling and ultimately in implementation. Fraunhofer IMM has the market advantage of being a pioneer in this research field, and the developed technology is already being requested and applied by industrial customers in bilateral cooperation projects within the framework of feasibility studies. The work of IMM on process implementation as well as on the reaction engineering conditions that lead to improved product quality is documented in several publications in peer-reviewed journals. (2 - 7) There are numerous activities in the field of chemical synthesis (especially in context of synthesizing fine chemicals and pharmaceuticals) to develop and utilize digital technologies. (8) The aspect of chemical process development is not excluded from this. This development coincides with the implementation of continuous manufacturing technologies in chemical synthesis, especially in the pharmaceutical industry. Automated Flow Chemistry reactor platforms with implemented real-time process analytical technology (PAT) play a central role in chemical process development, forming the

	basis for self-optimization, automatic development of reaction models, dynamic experiments, and the implementation of advanced process control strategies. (9, 10) Thus, the platforms move away from deterministic approaches with planned tasks and protocols towards greater autonomy through intelligent algorithms that enable the system to learn, make decisions, and implement them. Fraunhofer IMM has a range of own activities in this field. So, Fraunhofer IMM in cooperation investigated organometallic reactions to create a process model from the data obtained. This model allows for the prediction of optimal process parameters and process control for such compounds, despite remaining uncertainties in the data and incomplete model knowledge. This model (11) is now implemented as an optimization approach into an automated self-optimization lab set-up. Other activities are grouped around the automation of reactor and reaction modules for pilot and production scale in accordance with the new standard of the chemical industry (MTP: Module Type Package). This new standard is intended to allow individual, predefined modules of a complete plant to be integrated into an overall (automation) concept via Plug&Produce. The interaction with a central process control level, up to and including the process control system, plays a crucial role in ensuring successful integration. Therefore, activities are under way that the methods and algorithms of self-optimizing laboratory plant optimization should also be transferable to near-industrial automation solutions for pilot scale. Other works of Fraunhofer IMM have also been dedicated to the implementation of PAT for self-optimizing lab systems. (12)
Expected R&R of Korean SME Partner	The Korean SME will bring in the chemical synthesis strategy for the EUV photoresist precursor and will jointly develop with Fraunhofer IMM from this a continuous synthesis procedure. It will investigate produced material in view of quality and evaluate the new continuous synthesis procedure in view of overall efficiency and suitability for industrial implementation. The Korean SME will jointly with Fraunhofer IMM develop the general demonstrator plant concept, prepare the implementation on site, and will run the validation trials with the demonstrator plant. Through the development of ultra-high pure EUV photoresist precursors using AI-driven continuous-flow reactions, Korean partner will be able to establish a safe, efficient, and environmentally friendly manufacturing platform. This platform will minimize human toxicity and environmental hazards, while ensuring sustainability through high atom economy and low waste generation. By developing a novel process platform that integrates continuous-flow reactions, AI, and automation, Korean partner will be able to establish a scalable, manufacturing-ready technology for the mass production of ultra-high-purity EUV resist precursors. This approach enables commercialization that is not achievable with conventional batch-based processes. By integrating AI-based, data-driven process control, the Korean partner will enable significant improvements in quality and productivity, supporting the establishment of a strong global competitive advantage in the EUV photoresist market, as well as in advanced chip manufacturing for artificial intelligence (AI), data centers, and high-performance computing.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The following outcomes and impacts are expected from Korean SME and market perspective. **Expected Technical Outcomes:** • This project will establish highly pure EUV resist precursor manufacturing technology satisfying the stringent requirements of high-NA EUV lithography processes. • By integrating continuous-flow reactions with AI-driven automated process optimization, the project will achieve 1) enhanced metal ion control (from ppb to ppt levels) compared to conventional batch processes 2) reduction of organic byproducts, fine particles, and outgassing, and 3) minimized lot-to-lot quality variation, ensuring a high-reliability mass-production process for high-NA EUV resists. • These advancements provide Korean companies with the technological foundation to independently design and manufacture high-NA EUV resist precursors, reducing reliance on Japan- and U.S.-monopoly supply chains and enhancing material technology sovereignty. **Expected Economic Impact** • High-NA EUV resists will command 2–3 times higher prices than conventional EUV photoresists, with the market expected to grow at a CAGR of over 20% from 2027 to 2035. • Currently, the market is dominated by a few companies in Japan and the U.S. (e.g., JSR, TOK, DuPont), leaving Korean fabs with virtually no alternative sources. • Domestic Korean production of highly pure EUV resists is projected to secure 15–20% of the domestic fab supply market, resulting in an annual import substitution effect of approximately KRW 100 billion, and a global market share exceeding 5% in the high-NA EUV segment. • Introduction of AI- and flow-based automated manufacturing processes is expected to reduce production costs by 20–30% compared to conventional batch processes, improving price competitiveness, profitability, and incentivizing entry into new business areas. **Expected Employment Impact** • The AI- and flow-based advanced manufacturing platform is anticipated to create high-value-added technical jobs beyond routine manufacturing, including:1) R&D experts in reaction engineering and chemical process development, 2) process engineers skilled in data analysis and machine learning, and 3) specialists in equipment and system operation. • During scale-up and mass-production stages, approximately 20–40 skilled positions are expected, covering material synthesis/purification line operations, AI-based quality control (QC) system management, and customer fab technical support.

	• In particular, the workforce trained in AI-Flow integrated manufacturing will meet growing demand across the semiconductor, bio, and precision materials industries, contributing to the expansion of a robust industrial talent ecosystem. From Fraunhofer IMM perspective, the project generally will foster the further development of its technology basis of AI assisted Flow Chemistry and will strengthen its position as industry-oriented research and development service provider as its core business.

6. References

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X Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization